

The Evolving Urban Commute

Understanding Temporal Shifts in Trip Purposes in
the Puget Sound region

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Motivation

- **Travel behavior** studies often focus on **mode choice** and its **determinants**.
- Advances in **remote technologies** and **changes** in the **built environment** are **shifting** the **distribution of trip purposes**.
- While **some trip types** that are often **walked** may **decline**, increases in **walking for other purposes** could **offset this shift**, especially in response to built environment changes.
- Understanding these differences can help inform **better policies** to reduce car travel.



Commute



Shopping



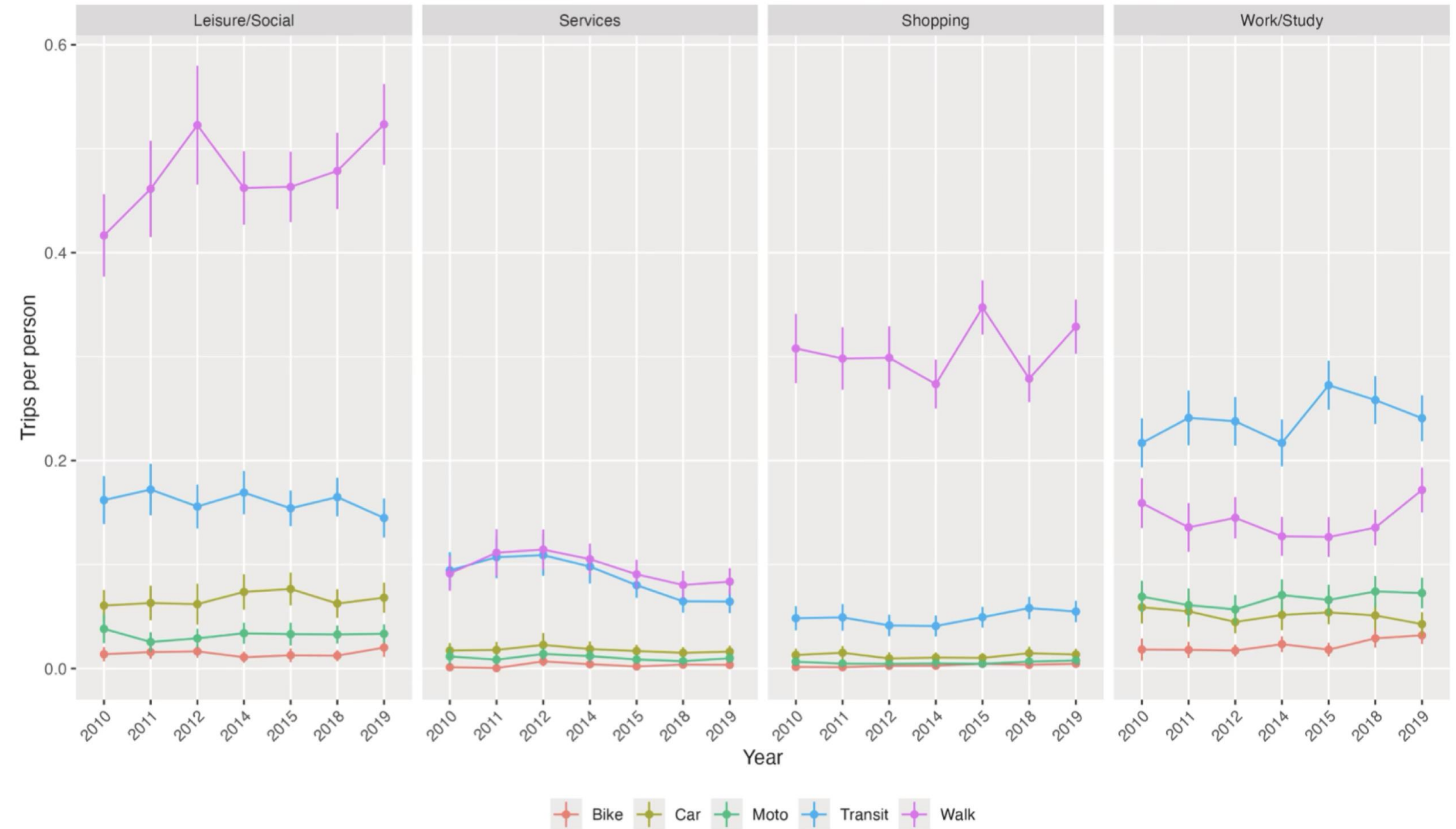
Social



Leisure

Motivation example

- Case of Barcelona based on yearly Regional Travel Survey (2010–2019)
- Shifts in trips per person over time



Analyzing Changes in Household Trip Purpose Rates Over Time

King County (Seattle), 2017–2019-2023 | Regional Household Travel Surveys

Goal: Detect shifts in household trip rates by purpose over multiple years

Data: Household travel surveys in the same region over multiple years

Two spatial strategies for temporal analysis (for household trips rates):

1. Census tract-level models (non-panel)

- Inconsistent tract coverage limits longitudinal tracking
- Include year dummies in pooled models to test time effects

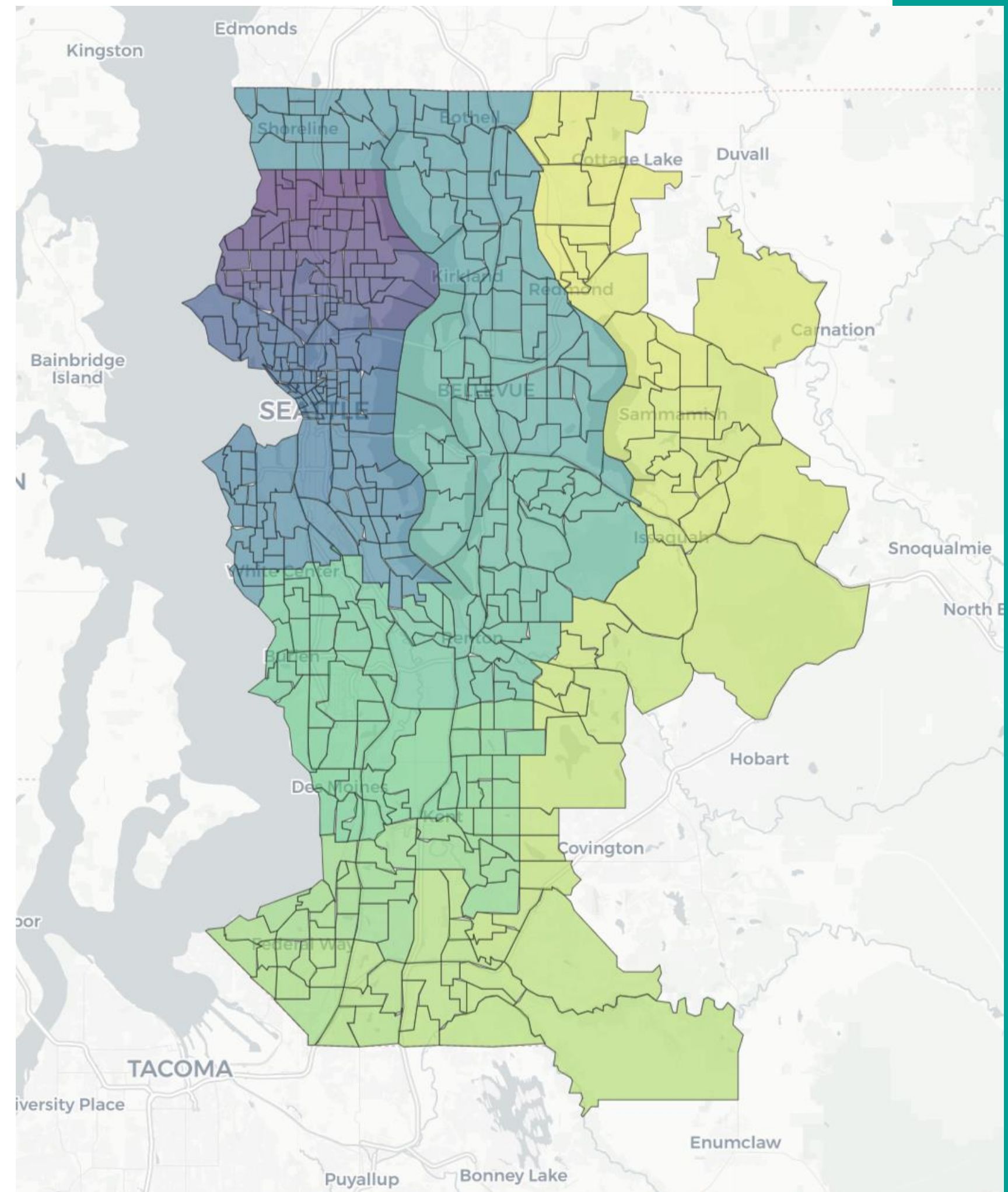
2. Spatial clustering (pseudo-panel approach)

- Aggregate tracts into stable spatial units (clusters)
 - Construct area-level panel for fixed-effects models over time
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King County Seattle

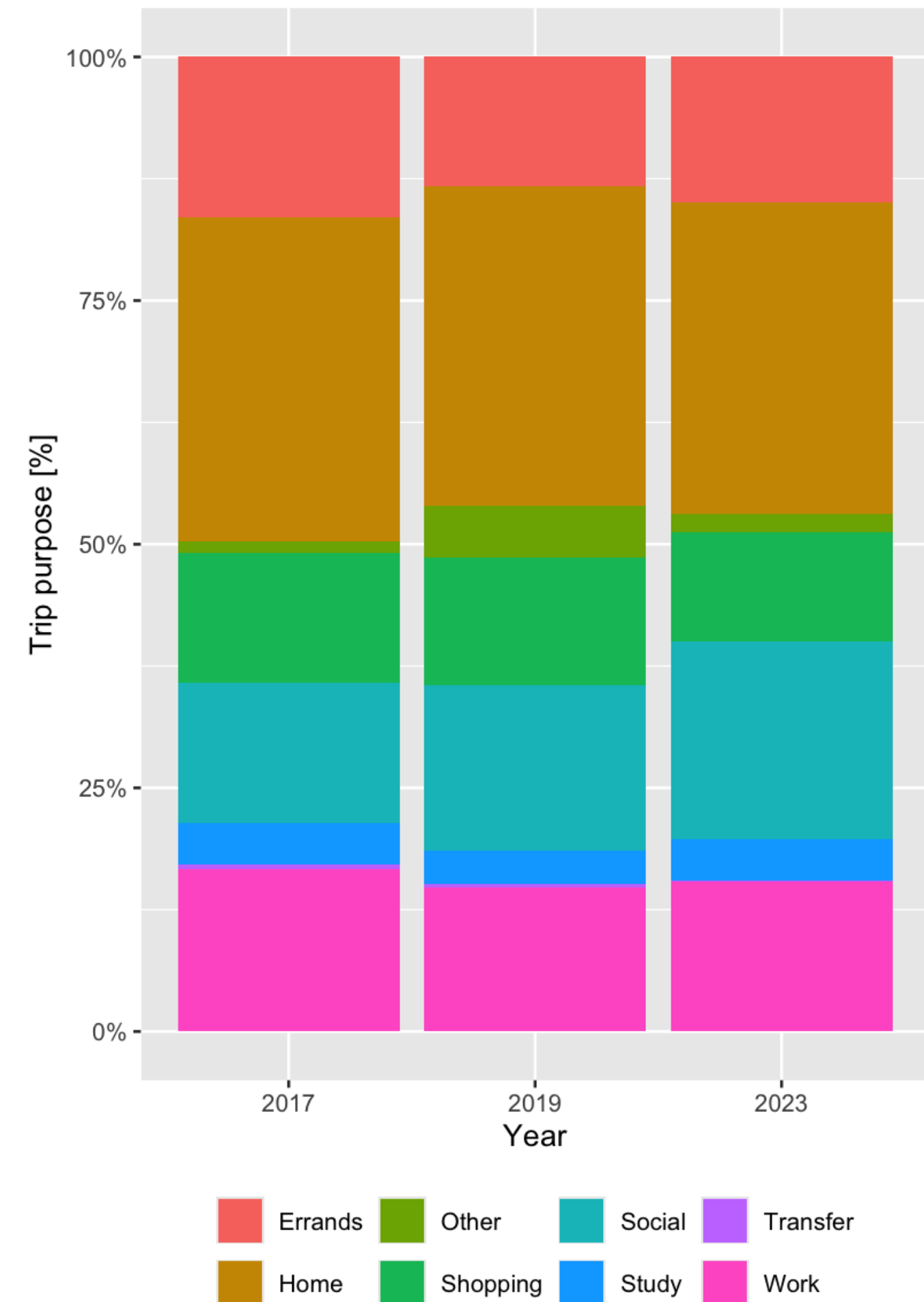
- Center of the Puget Sound Region
 - ~2 million residents across ~1,600 km²
 - 7% increase in population between 2017-2023
 - Region was selected due to the availability of household travel surveys for multiple years with the aim of testing methods
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Trip purpose distribution

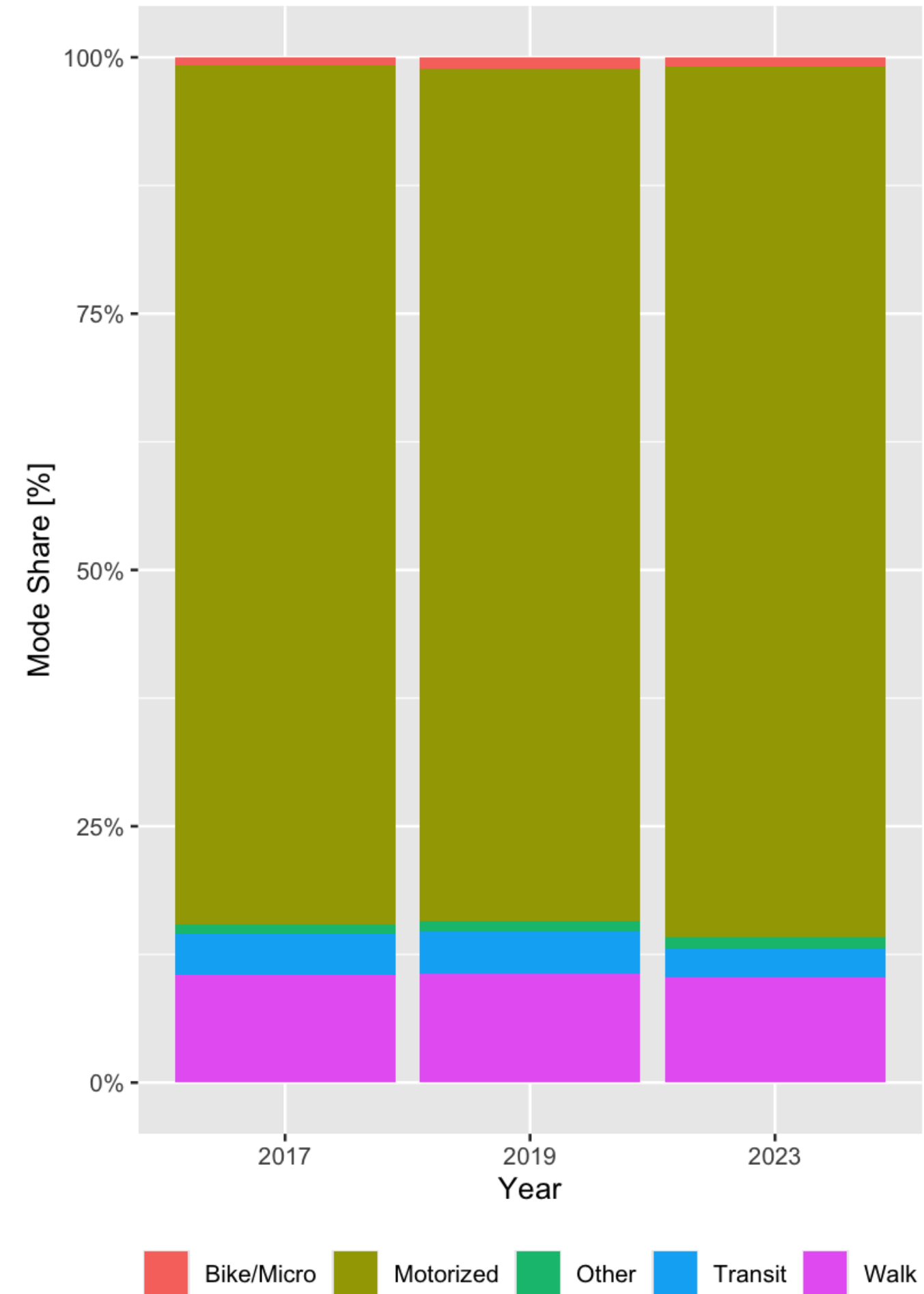
- Shopping: steady decline
- Errand: decline followed by rebound
- Social: gradual increase
- Work: overall decline





Mode Share

- Overall modal share remained relatively stable
- Decline transit use





Household Models

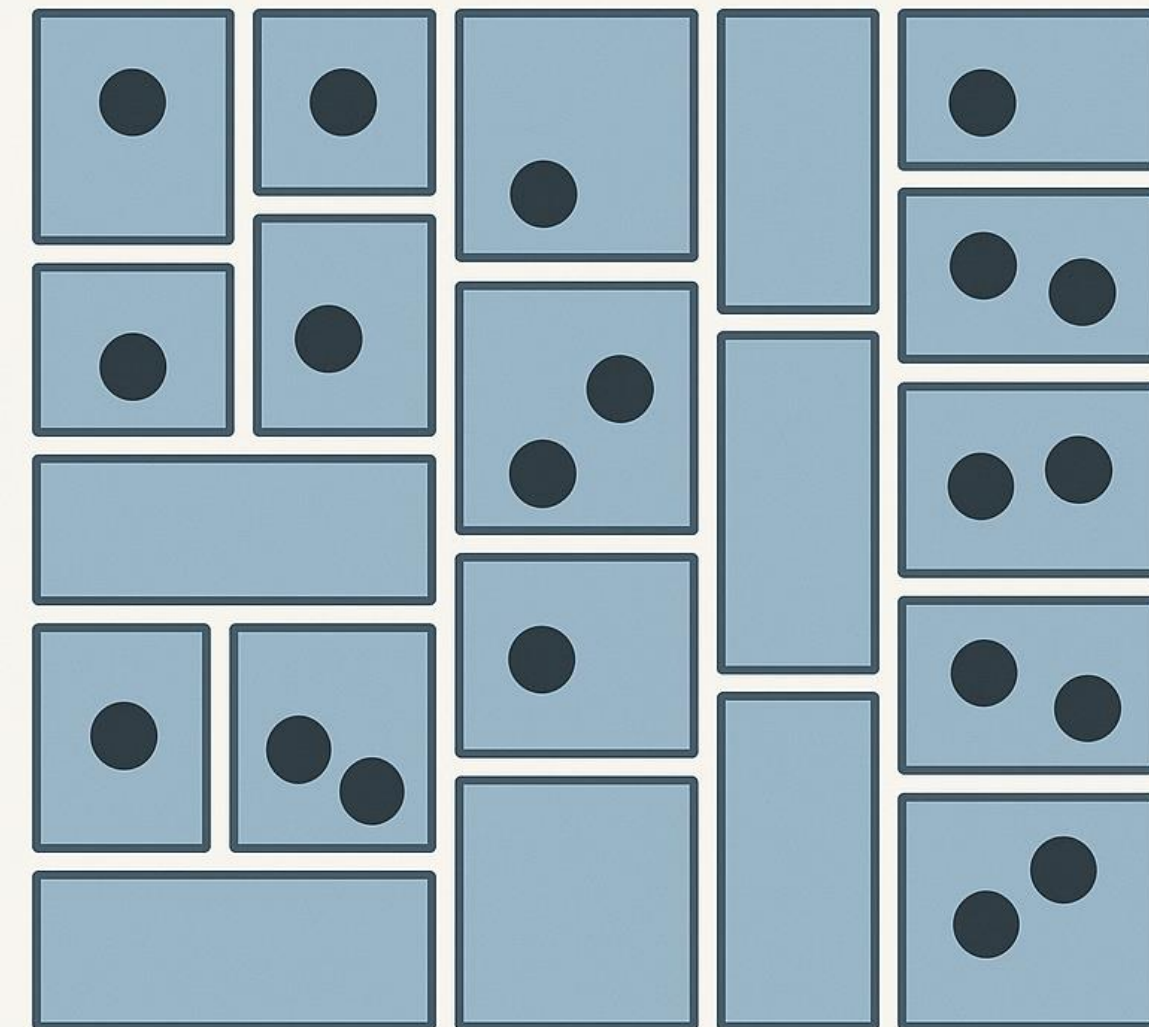
Independent variable

- **Household trip rate by purpose**
Errands, Shopping, Social, Work

*Dependent variables at **census tract level***

- **Sociodemographic characteristics**
- **Transit accessibility**
- **Local environment** measures (population density, jobs density)
- **Year**

Linear regressions for each purpose



**MODELING AT THE
HOUSEHOLD LEVEL**

Spatial Cluster Pseudo Panel Models

Zones aggregated into spatial clusters (from the census tracts) using maxp method (aggregating similar areas)

Independent variable

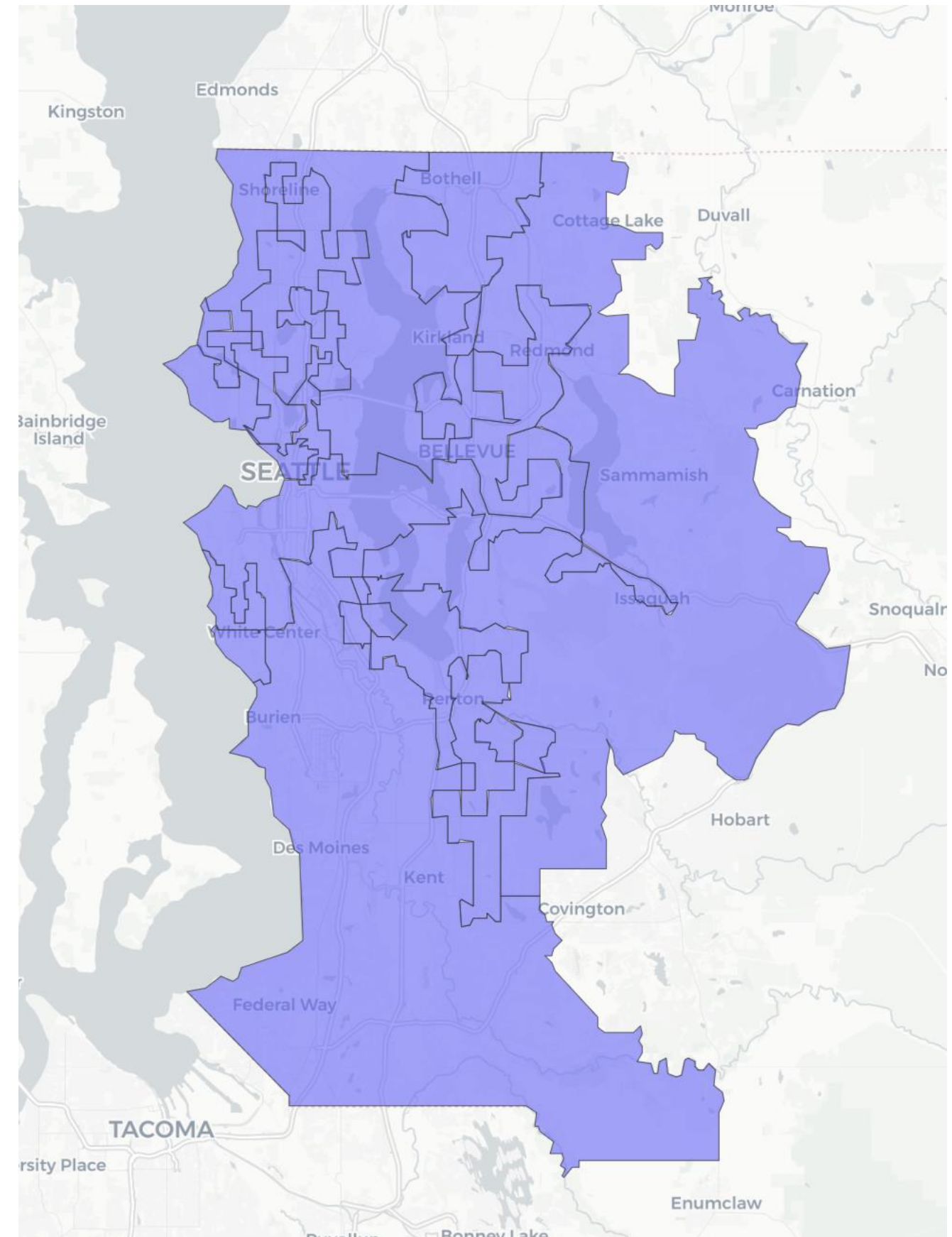
- **Mean household trip rate by purpose inside cluster**

Errands, Shopping, Social, Work

*Dependent variables at the **cluster level***

- **Transit accessibility**
- **Local environment** measures (population density, jobs density)
- **Year**

Linear regressions for each purpose





Results household models

- **Errands** show a partial recovery from the decline observed in 2019.
- **Shopping** trips decline clearly in 2023 compared to 2019.
- **Social** trips increase slightly over time, but changes are not statistically significant.
- **Work** trips exhibit a steady linear decrease across the years.
- **Model performance** is weak, with low explanatory power.
- **Built environment variables** have limited influence on explaining trip rates.

Variable	Errands	Shopping	Social	Work	Total
(Intercept)	-1.245	1.079	0.460	1.450	1.482
Year: 2019	-0.065	-0.041	0.198	-0.386	-0.078
Year: 2023	-0.034	-0.216	0.224	-0.575	-0.607
hhsiz: 2	0.504	0.315	0.976	0.365	2.446
hhsiz: 3+	2.018	0.799	1.259	0.043	5.583
vehicle: 0	-0.654	-0.433	-1.033	-0.567	-2.732
vehicle: 1	-0.207	-0.332	-0.305	-0.079	-0.830
workers: 0	0.961	1.294	0.557	-2.104	0.989
workers: 1	0.450	0.398	0.288	-0.806	0.604
Jobs in 30min transit (100,000s)	-0.021	0.141	0.094	-0.090	-0.727
Population in 30min transit (100,000s)	0.050	-0.307	0.290	0.002	0.258
Jobs density (100 jobs/ha)	-0.003	-0.013	0.022	0.026	0.037
Population densuty (100 inhabitats/ha)	0.452	-0.059	0.177	0.510	1.217
Median income (USD 1000s)	0.003	0.004	0.041	-0.026	0.003
Median age	0.036	-0.013	-0.006	0.033	0.046
R2	0.030	0.016	0.017	0.085	0.031

Bold p<0.1



Results Spatial Clusters models

- **Errands, Shopping, and Work** show a steady decline in trips per household
- **Shopping** has the best model fit overall; in 2023, the drop in shopping rates is twice as large as in 2019
- **Social** trips have the weakest model, with no statistically significant predictors
- **Model performance** improves overall, likely due to aggregation, which reduces noise—as expected
- **Built environment variables** again show limited and inconsistent influence on trip rates across models.

Variable	Errands	Shopping	Social	Work	Total
(Intercept)	0.5	-1.4	1.3	1.5	7.8
Year: 2019	-0.3	-0.4	-0.3	-0.2	-1.7
Year: 2023	-1.0	-0.7	-0.3	-0.4	-3.5
Jobs in 30min transit (100,000s)	-0.3	-0.2	-0.6	-0.3	-1.6
Population in 30min transit (100,000s)	-2.1	0.0	0.6	0.0	-2.5
Jobs density (100 jobs/ha)	0.2	0.0	0.1	0.0	0.3
Population density (100 inhabitants/ha)	1.6	0.5	0.5	0.5	3.2
Median income (USD 1000s)	0.0	0.0	0.0	0.0	0.0
Median age	0.1	0.1	0.0	0.0	0.1
R2	0.4	0.8	0.2	0.4	0.5

Bold p<0.1



Discussion

- This is an initial and exploratory attempt. There are important limitations (models are instable, low statistical power)
- Thoughts?

Next Steps

- **Explore methods to detect substitution effects**, such as ecological inference.
 - **Refine model specification** to better capture behavioral dynamics.
 - **Refine the link between mode choice and trip purpose**, identifying whether the built environment influences both dimensions or interacts differently with each.
 - **Seek additional case study cities** with more balanced modal splits (e.g., Barcelona), contingent on obtaining more detailed or disaggregated data.
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Thanks!
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